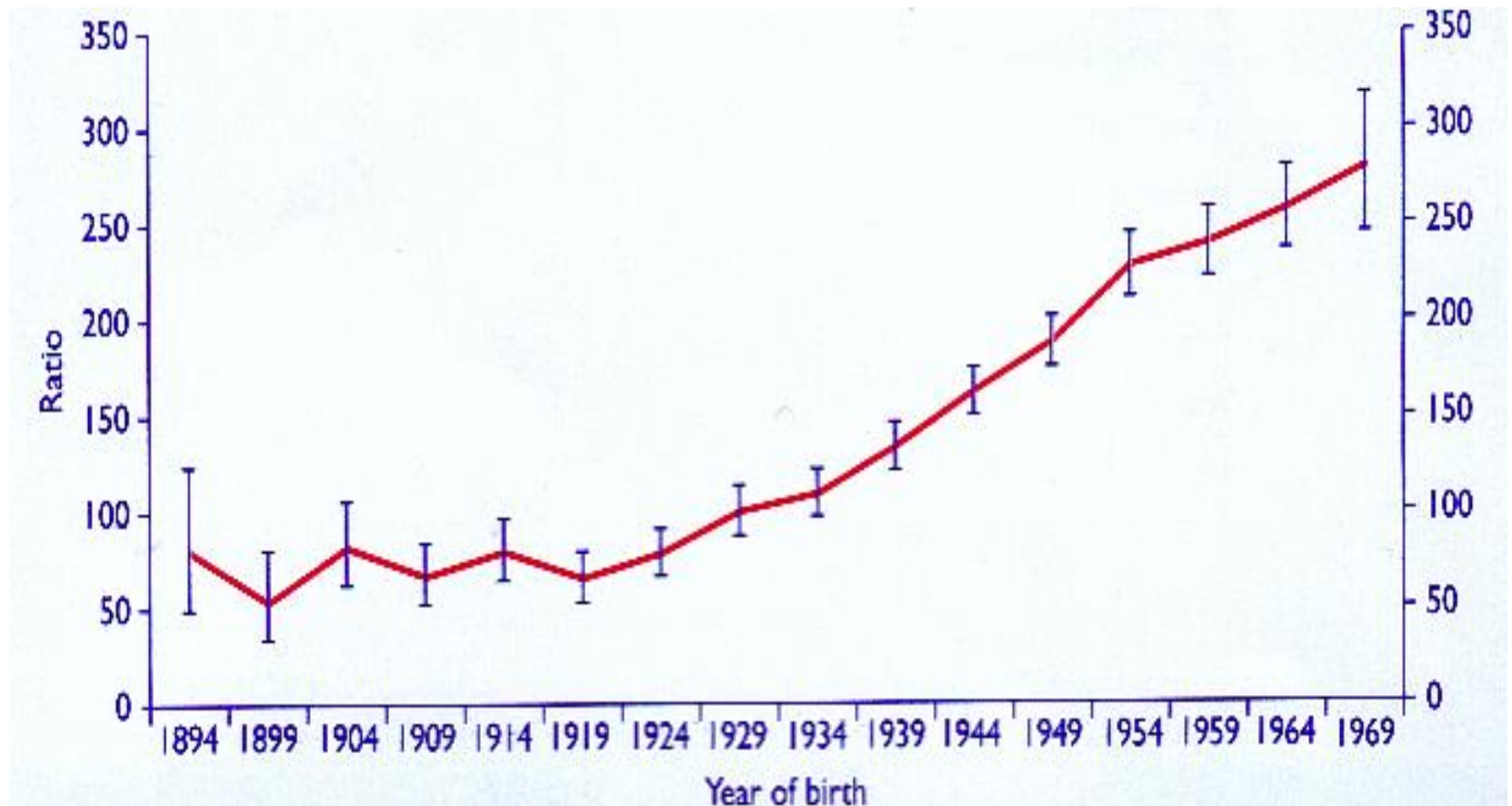


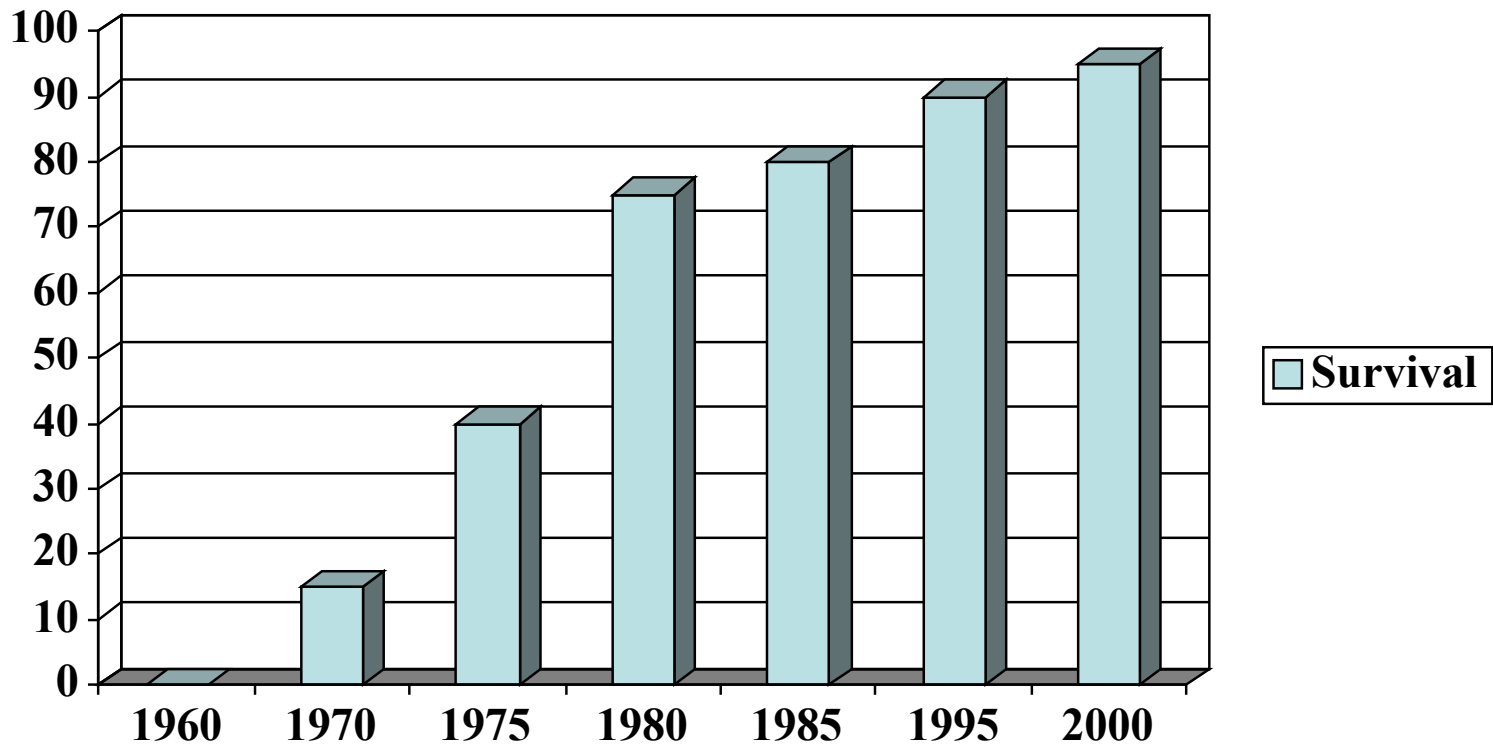
Germ Cell Cancers 2014

Alan.Horwich@icr.ac.uk
Royal Marsden Hospital UK

Incidence of TGCT by birth cohort



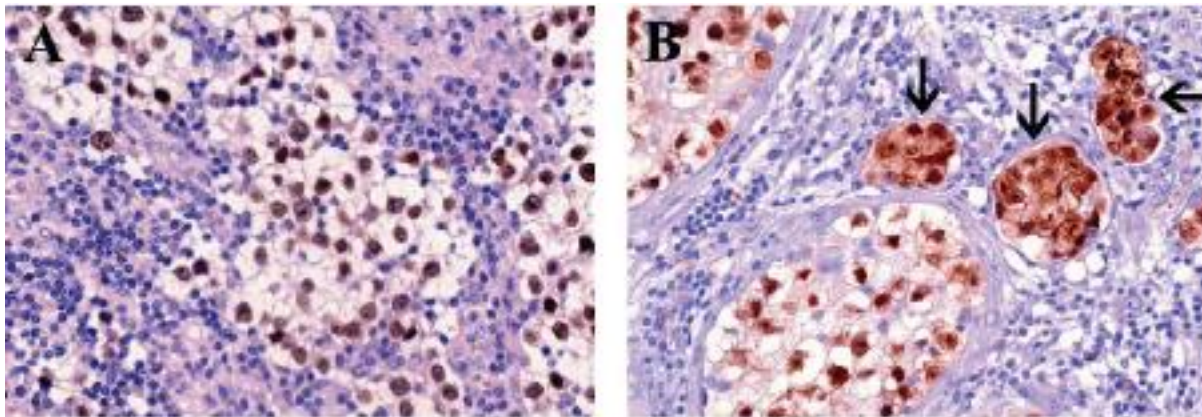
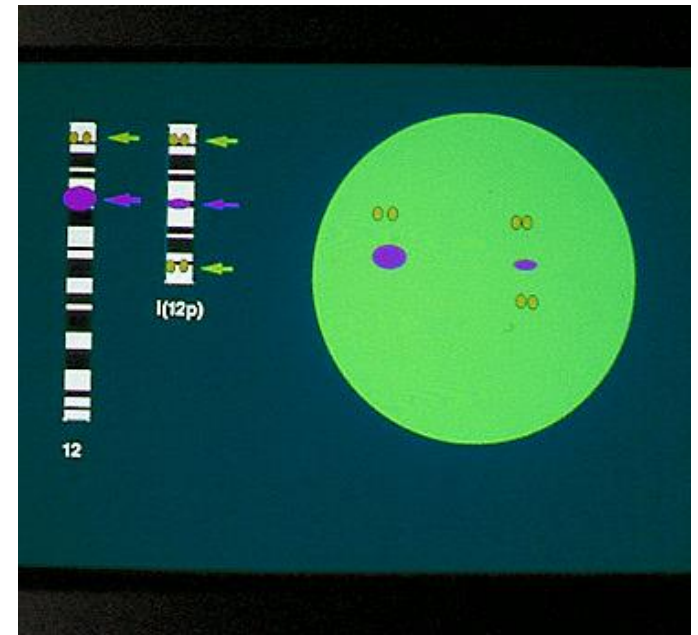
Survival of good prognosis metastatic GCT



Advances

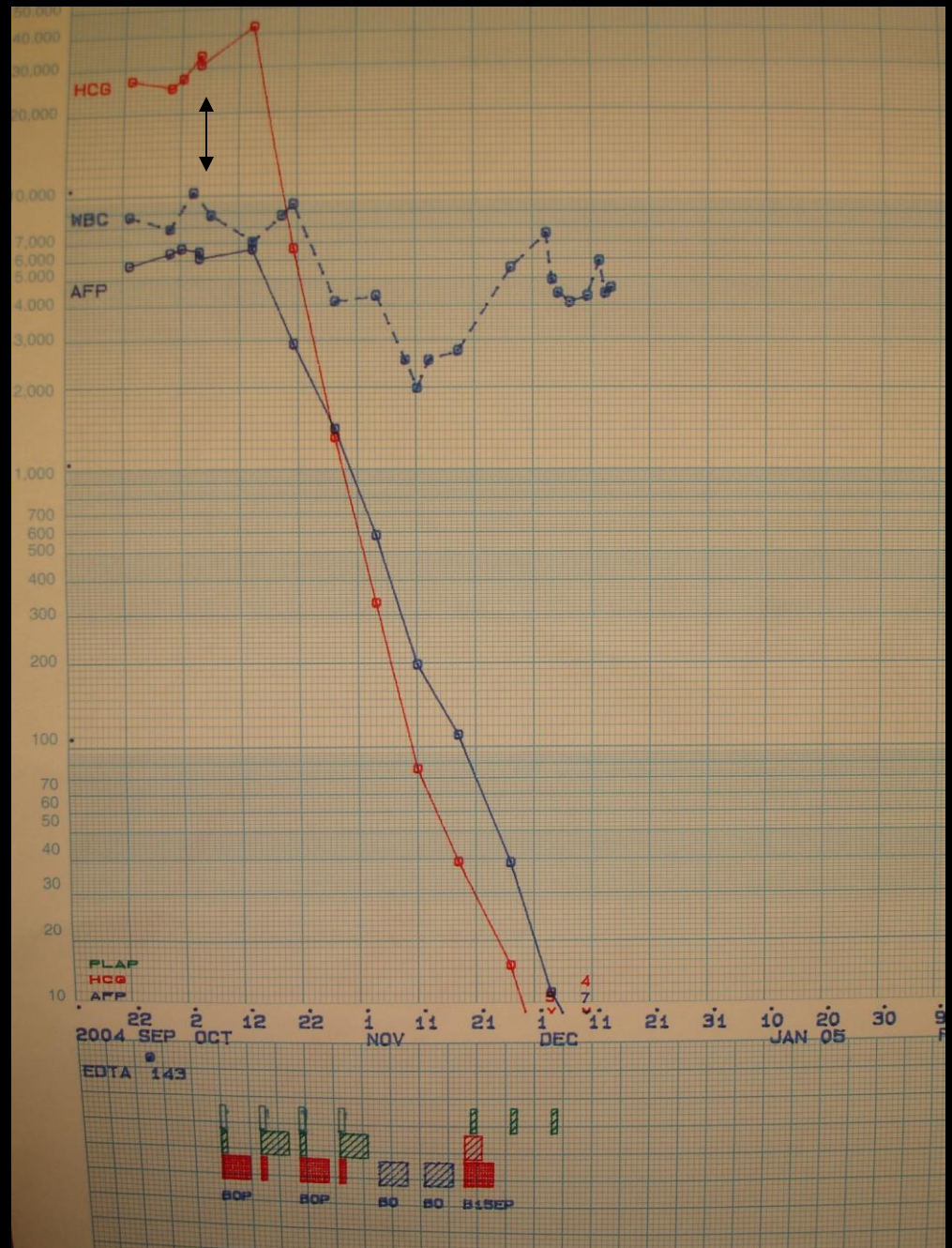
- Diagnosis/imaging
- Surgery
- Radiotherapy
- Chemotherapy
- Monitoring
- Late toxicity risks

80% of male GCT contain an
isochromosome 12p
[i(12p)]

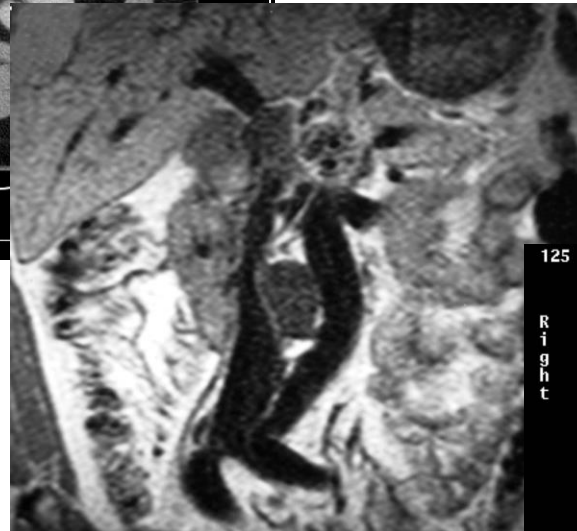
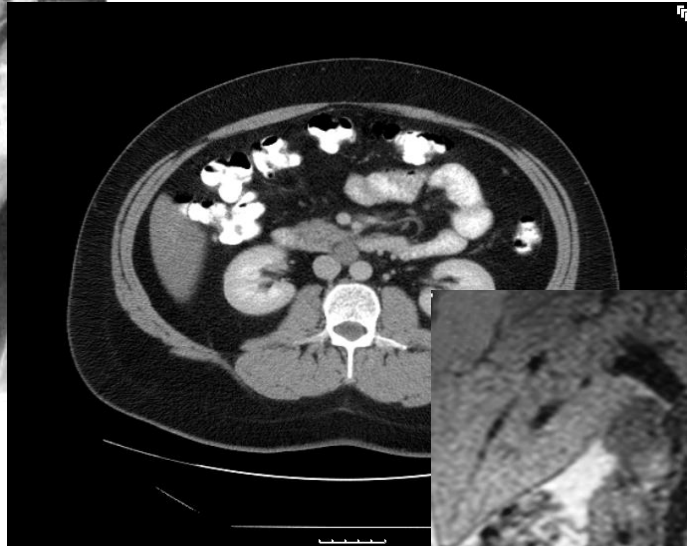


OCT3/4 staining in (A)seminoma (B) embryonal carcinoma
with seminoma

Marker surge
after start of
chemotherapy



TT Imaging



Role of surgery in advanced nonseminoma

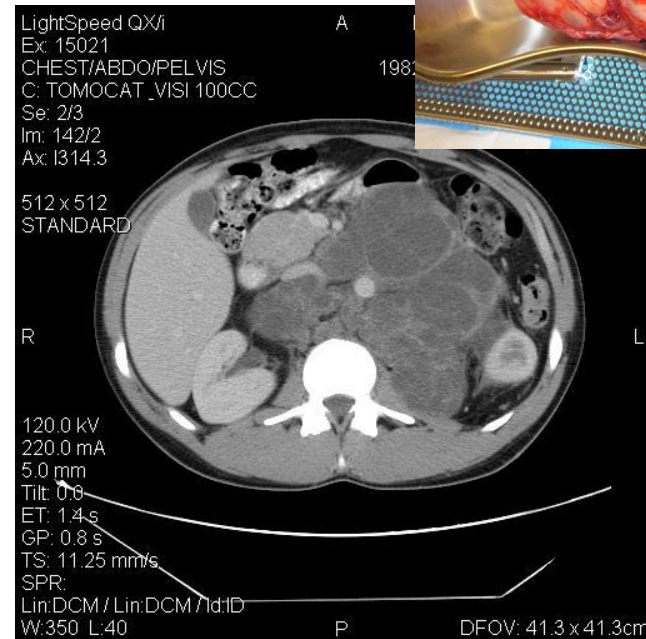
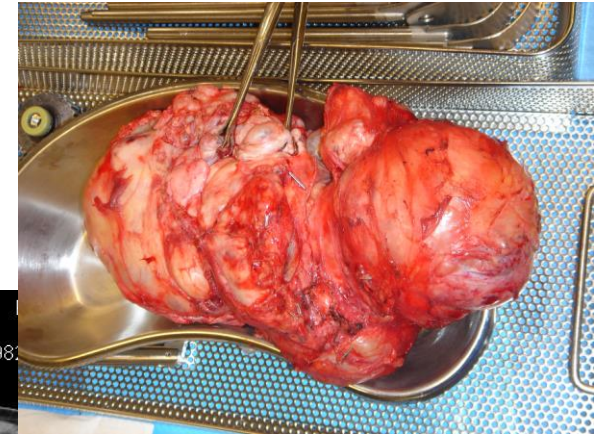
Surgical excision should be initial treatment if feasible for:

Residual mass after chemotherapy

Growing teratoma

Late relapse

Transformation of teratoma



Germ cell tumours Stage at presentation RMH '94- ' 03

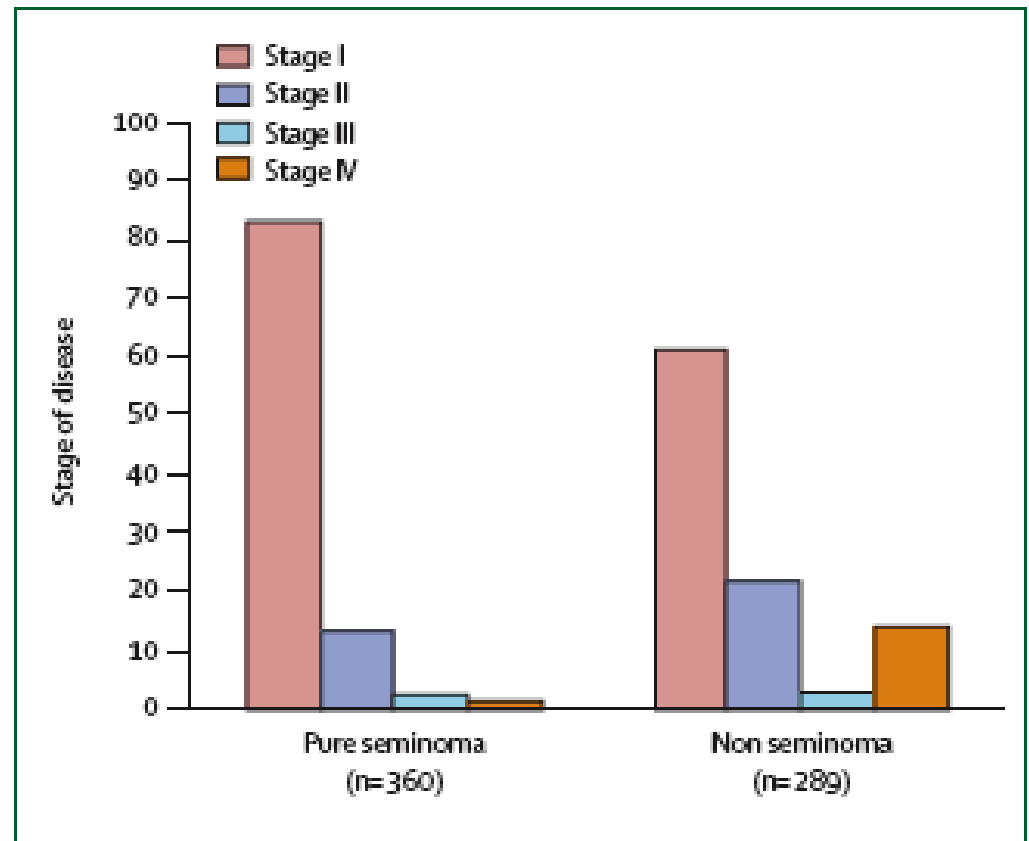


Figure 3: Proportion of patients with stages of seminoma and non-seminoma (1994–2003)

Data are from the Royal Marsden Hospital Records.

From Testicular germ-cell cancer

Horwich Shipley Huddart Lancet 2006 367 754

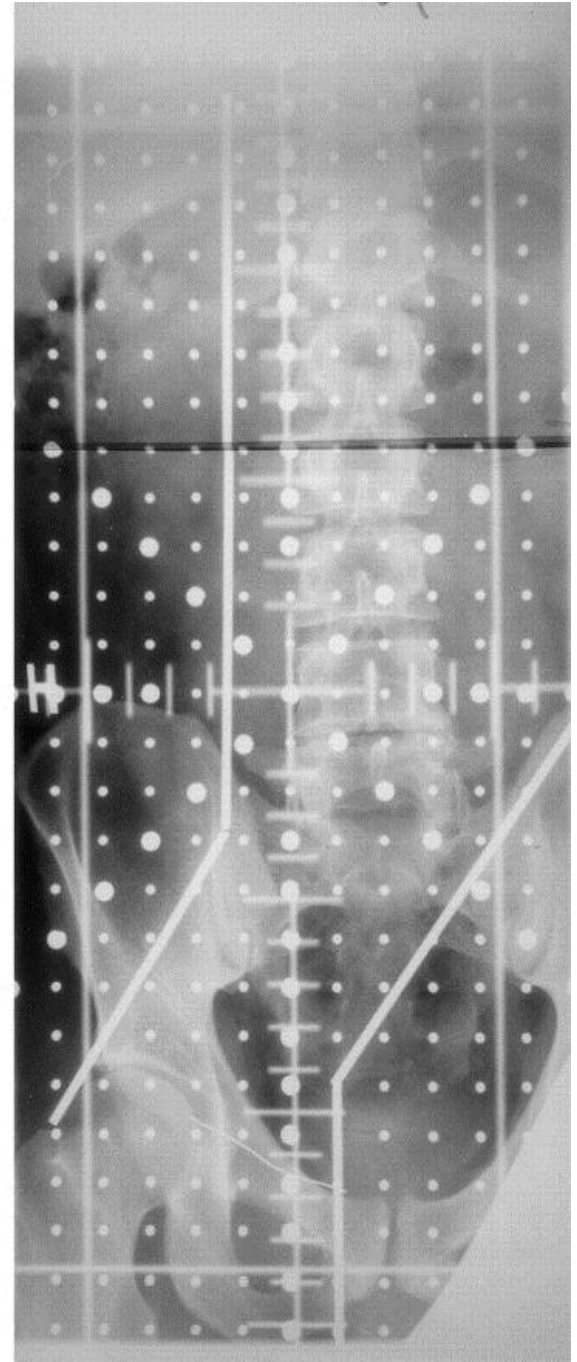
Historical role of RT in St 1 Seminoma

- Very radiosensitive tumour
- Predictable sites of initial spread
- Traditional treatment thus long-term outcomes known

BUT

- Young patients so future risks are important

2014-RT is rarely used in TTs

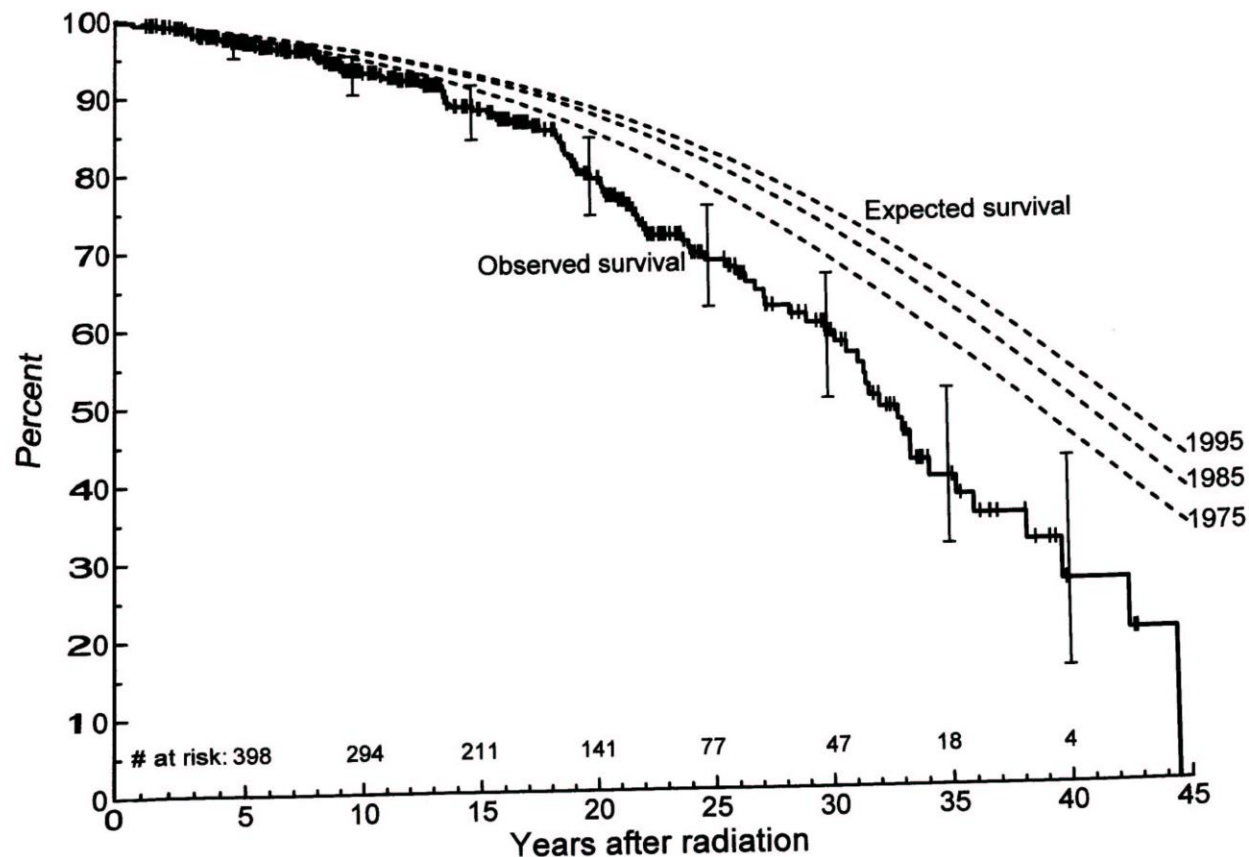


MDA survivorship analysis

Zagars et al 04

Stage 1 and 2 seminoma RT alone N=453

Survival post RT compared to US matched population

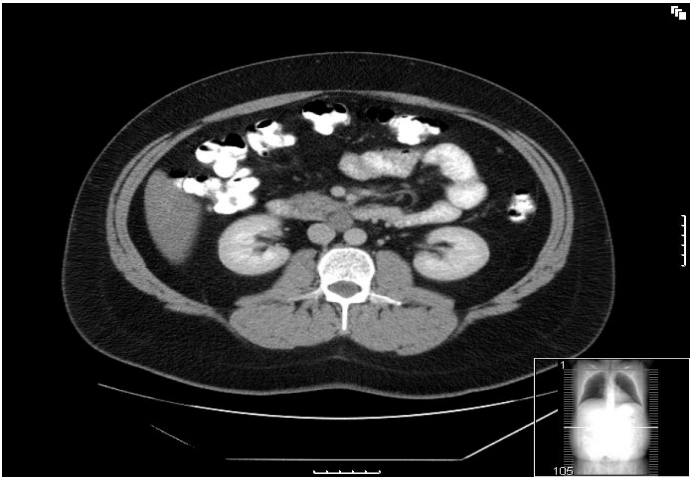


Seminoma - Surveillance Follow-up Schedule

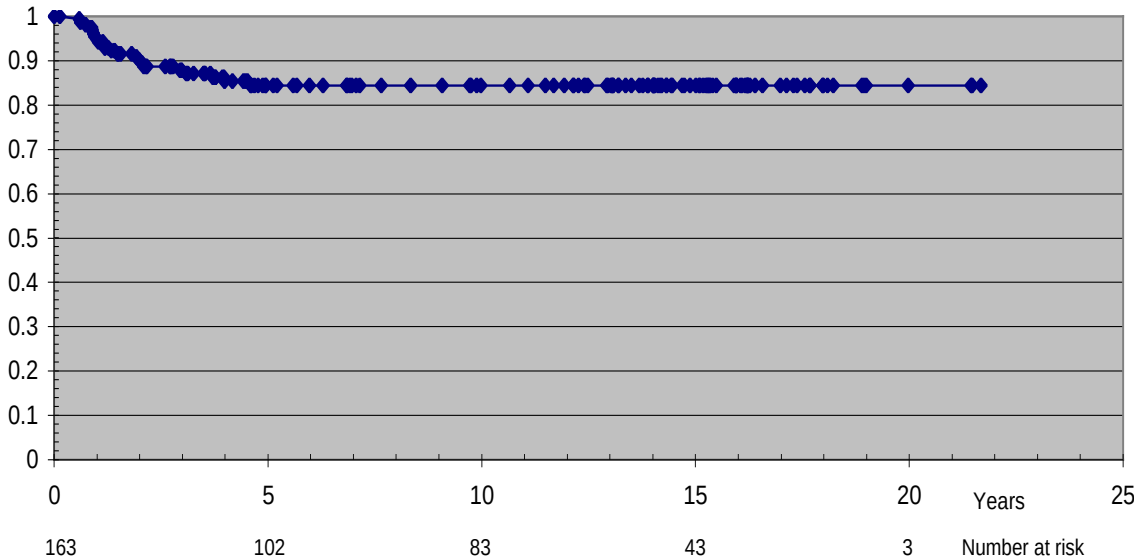
Name: A Patient
Orchidectomy date: xx/01/2001

Month	Clinical	Markers	CXR	CT abdomen
3	O	O		
6	O	O	O	O
9	O	O		
12	O	O	O	O
15	O	O		
18	O	O	O	O
21	O	O		
24	O	O	O	O
28	O	O		
32	O	O		
36	O	O	O	O
42	O	O		
48	O	O	O	O
54	O	O		
60	O	O	O	O
72	O	O		
84	O	O		
96	O	O		
108	O	O		
120	O	O		

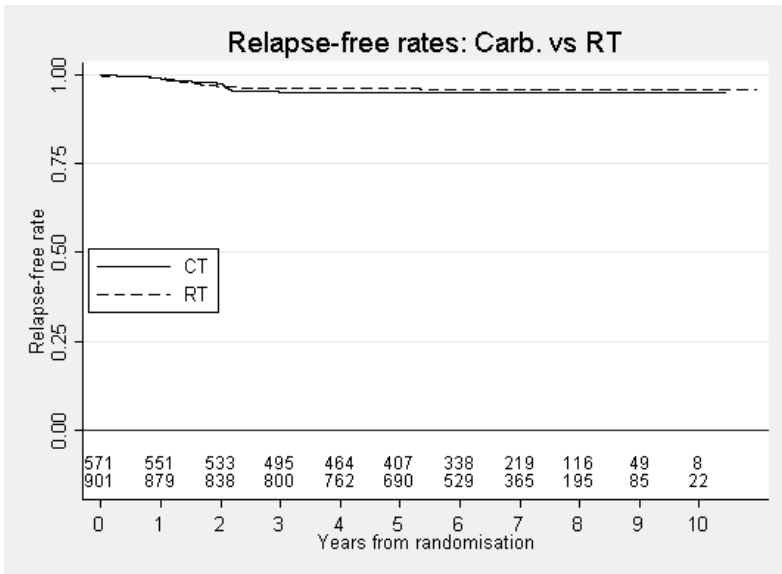
CT scans should be abdomen only unless pelvis at risk
Markers: AFP, β -HCG, LDH



Stage I Seminoma Surveillance - Relapse Free Rate



Results: Overall relapse-free rate



RFR at 5 years (95% CI)

R 96.0% (94.5%, 97.1%)

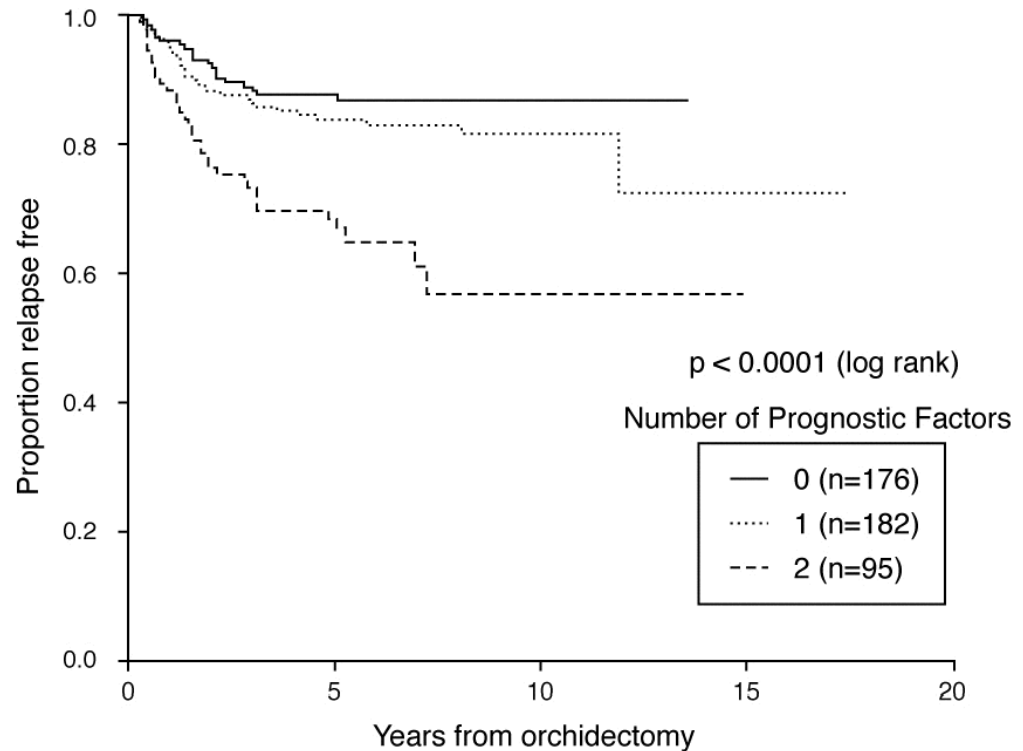
C 94.7% (92.5%, 96.3%)

**TD Oliver, GM Mead, PJ
Fogarty and SP Stenning
for the MRC/EORTC
collaborators**

**Radiotherapy vs carboplatin for stage I
seminoma**

Seminoma stage 1: risk factors for relapse on surveillance

- Diameter > 4cm
- Rete testis invasion



Risk Factors

5 year
recurrence rate

0

12.2%

1

15.9%

2

31.5%

Changes in management of stage I seminoma

- Adjuvant radiotherapy
 - only para-aortic field
 - Only 20Gy/10 fractions/2 weeks
 - radiotherapy now not used in the UK
- Alternatives
 - Surveillance
 - defined protocol
 - 5 year follow up with cross sectional imaging
 - Carboplatin-1 cycle

Chemotherapy regimens for testicular tumours

FIRST REPORT OF BEP AS INITIAL CHEMOTHERAPY

The treatment of metastatic germ-cell testicular tumours with bleomycin, etoposide and cis-platin (BEP)

M.J. Peckham, A. Barrett, K.H. Liew, A. Horwich, B. Robinson, H.J. Dobbs, T.J. McElwain & W.F. Hendry
Br. J. Cancer (1983), 47, 613-619

Institute of Cancer Research and The Royal Marsden Hospital London & Surrey.

BEP

Cisplatin	20mg/m ² /day	days 1 - 5
Etoposide	100mg/m ² /day	days 1 - 5
Bleomycin	30, 000iu	days 2, 8, 15

Surveillance post-orchidectomy for Stage I **Nonseminoma**

30 % patients relapse overall OS 98-99 %
40-50% relapse if vascular invasion and 15-
20% if not

Advantages of surveillance

Avoids further treatment in 70 % patients

Disadvantages

3/4 cycles chemo if relapse **and 26% also need
RPLND**

Compliance

Adjuvant chemotherapy in stage I nonseminoma

In high risk patients reduces RR 50 % to <5%

UK MRC Trial based on 2 x BEP (Etoposide 360 mg/m²) in 114 high risk patients. At median 4 years there was 1 GCT relapse and 1 carcinoma of the rete testis.

No loss of fertility from 9 months after chemotherapy

2 studies suggest a single cycle of BEP is effective,

Eg 1 SWENOTECA retrospective, found 3% RR in high risk.

Eg 2 German Trial vs RPLND

MRC prospective trial in 246 high risk patients awaited.

Randomized Phase III Trial Comparing Retroperitoneal Lymph Node Dissection With One Course of Bleomycin and Etoposide Plus Cisplatin Chemotherapy in the Adjuvant Treatment of Clinical Stage I Nonseminomatous Testicular Germ Cell Tumors: AUO Trial AH 01/94 by the German Testicular Cancer Study Group

Peter Albers, Roswitha Siener, Susanne Krege, Hans-Uwe Schmelz, Klaus-Peter Dieckmann, Axel Heidenreich, Peter Kwasny, Maik Pechoel, Jan Lehmann, Sabine Kliesch, Kai-Uwe Köhrmann, Rolf Fimmers, Lothar Weißbach, Volker Loy, Christian Wittekind, and Michael Hartmann

From the Department of Urology, Klinikum Kassel GmbH, Kassel; Department of Urology, Bonn University; Institute of

No=382

2 year recurrence-free rate

CT =99% S=92%

H.R. 7.9 p =.001

Therefore 1 cycle BEP as adjuvant

Also supported by SWENOTECA report
(Tandberg et al 2009)

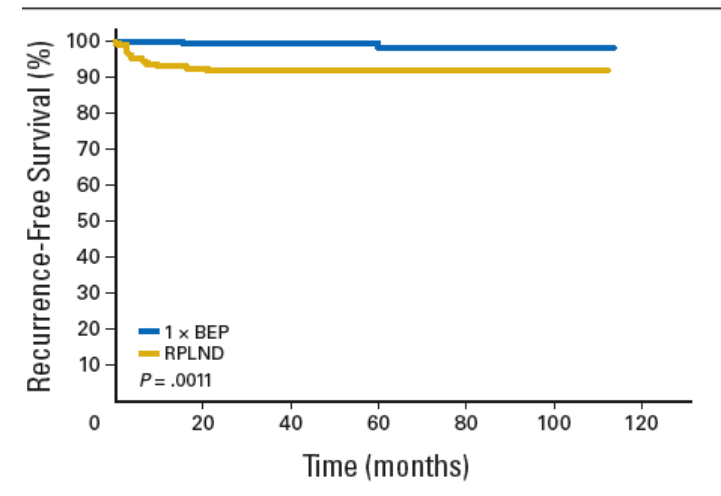


Fig 2. Recurrences in the intention-to-treat population. BEP, bleomycin, etoposide, and cisplatin; RPLND, retroperitoneal lymph node dissection.

PROGNOSIS of metastatic germ cell cancer

An initiative from the MRC TTWP 1994

IGCCCG classification for metastatic nonseminomas

Category	Criteria	HCG Level iu/l	AFP Level iu/l	LDH x normal	NPVM	Media- stinal primary	5yr %PFS
Good	All of	<5000	<1000	<1.5	no	no	89
Inter- mediate	All of	5-50,000	1-10,000	1.5-10	no	no	75
Poor	Any of	>50 000	>10 000	>10	Yes	Yes	41

NPVM=Non Pulmonary Visceral Metastasis eg liver, bone, brain

Long term symptomatic effects of BEP chemotherapy

- Peripheral neuropathy 20%
- Raynauds 22%
- Tinnitus 24%
- Hearing threshold 34db
- Unable to conceive 29%

INCREASED RISKS: Leukaemia and cardiovascular disease

MRC/EORTC trial: 3 v 4 cycles in good prognosis GCT

n = 812

Median FU 25 months

BEP x 3

BEP x 3 +EP x 1

2yr PFS

90.4%

89.4% (Diff 0.99%
CI 3.8- 1.83%)

CR rates

70.8%

73.4%

Relapses

24

23

De Wit et al 2001

Therefore only 3 cycles given for good prognosis
patients (the majority of those with metastases)

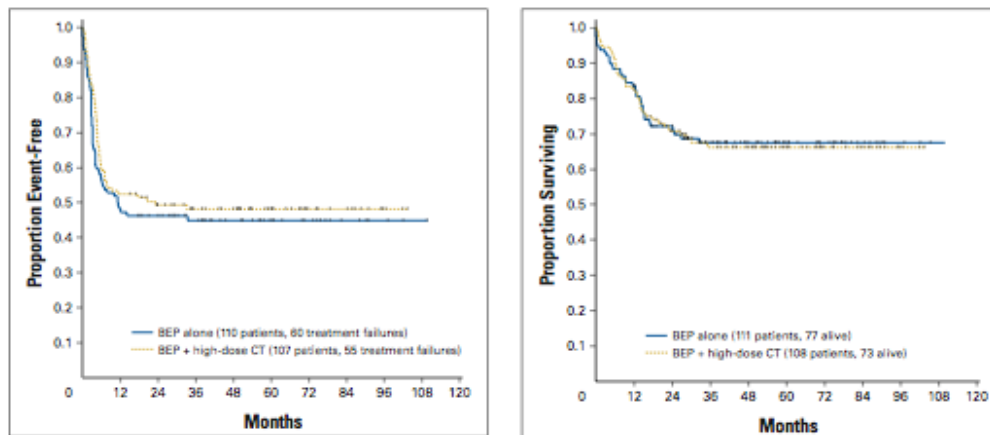
Phase III Randomized Trial of Conventional-Dose Chemotherapy With or Without High-Dose Chemotherapy and Autologous Hematopoietic Stem-Cell Rescue As First-Line Treatment for Patients With Poor-Prognosis Metastatic Germ Cell Tumors

Robert J. Motzer, Craig J. Nichols, Kim A. Margolin, Jennifer Bacik, Paul G. Richardson, Nicholas J. Vogelzang, Dean F. Bajorin, Primo N. Lara Jr, Lawrence Einhorn, Madhu Mazumdar, and George J. Bosl

4x BEP v 2x BEP + 2x High dose ChemoT

N= 219 Intermediate and poor risk

High-Dose Chemotherapy in Germ Cell Tumors



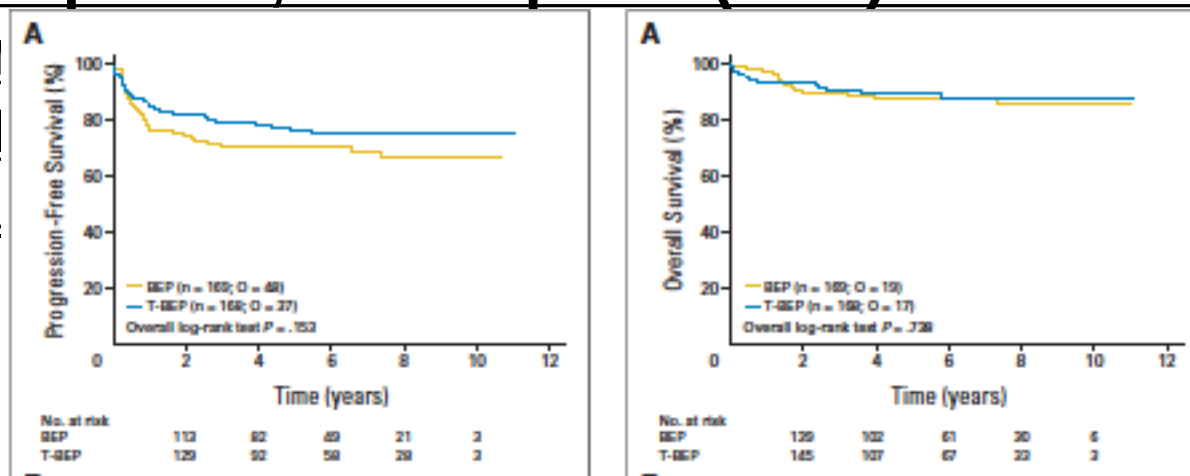
Therefore 4 cycles of BEP remain the standard approach

Adding Paclitaxel to BEP for Intermediate Prognosis

J Clin Oncol. 2012 Mar 10;30(8):792-9

Randomized phase III study comparing paclitaxel-bleomycin, etoposide, and cisplatin (BEP) to standard BEP in intermediate prognosis

BEP in intermediate prognosis
de Wit R, et al



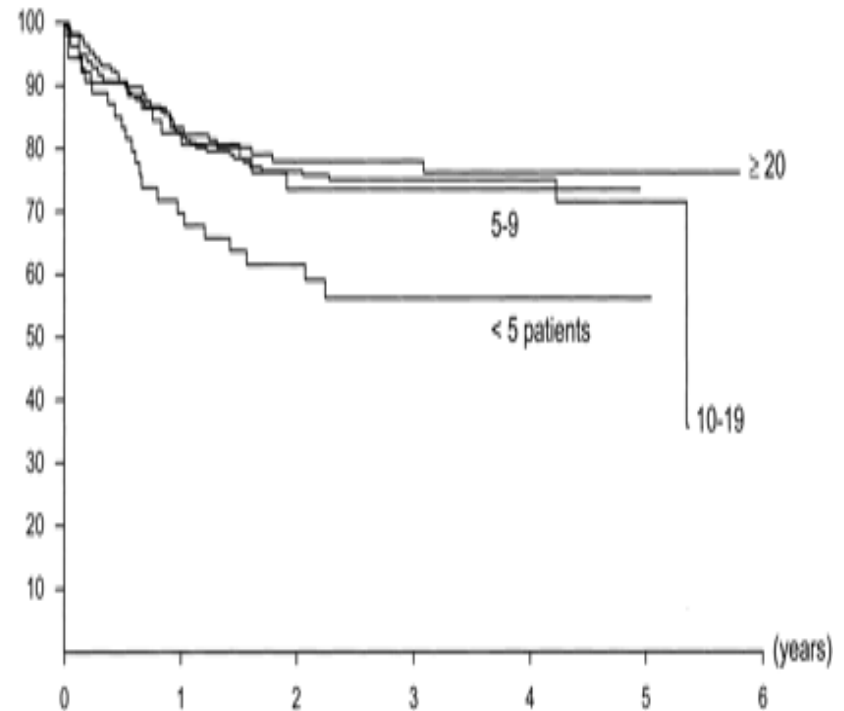
At 3 years PFS 79 vs 71% and Overall Survival 91 vs 89% on Intent to treat analysis n=337 (NS)

In analysis only of eligible patients (n=305), 3 yrs PFS was 83 vs 70%, p0.03 (OS no diff)

Effect of institution where patient was treated:

Analysis of BOP/VIP v BEP trial (Collette 1999)

- 380pts from 49 institutions in 11 countries
- Institutions entering <5 patients (n=26, 55pts)
 - had double the risk of death HR 1.85 p=0.0099
 - Similarly reduced PFS and FFS (HR1.56-1.85)
 - Lower CR rate 31% v 43%



Survival inferior in centers treating small number of high risk patients

Histology at post-chemotherapy disease resection

Author	N	Histology of mass (%)		
		Fibrosis/ necrosis	TD	MTU
Herr	122	47	39	14
Donohue	870	27	47	23
Hendry				
Elective	330	25	66	8
Salvage	112	13	38	49

TD= Teratoma Differentiated-possible long term hazard

MTU= persisting active cancer- a possible short term hazard

Residual malignant cancer after chemotherapy

Multicentre retrospective analysis of resected malignant disease post chemotherapy n=238

- Prognostic factors:
 - IGCCCG good,
<10% viable cells,
Complete resection
 - 0 100% PFS
 - 1 83% PFS
 - 2/3 40% PFS
- Intermediate risk factor - adj chemo improved PFS but not survival

Poor prognosis syndromes

1. Relapse after firstline chemotherapy

Salvage by second-line or high dose chemotherapy plus **surgery** if feasible

2. Transformation of teratoma

Surgery is critical for cure

3. Mediastinal primary.

Early **surgery** after chemotherapy.

Conventional Dose Salvage

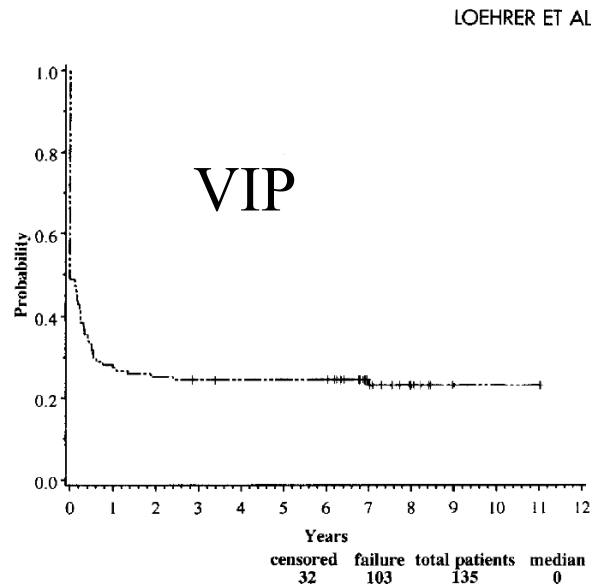
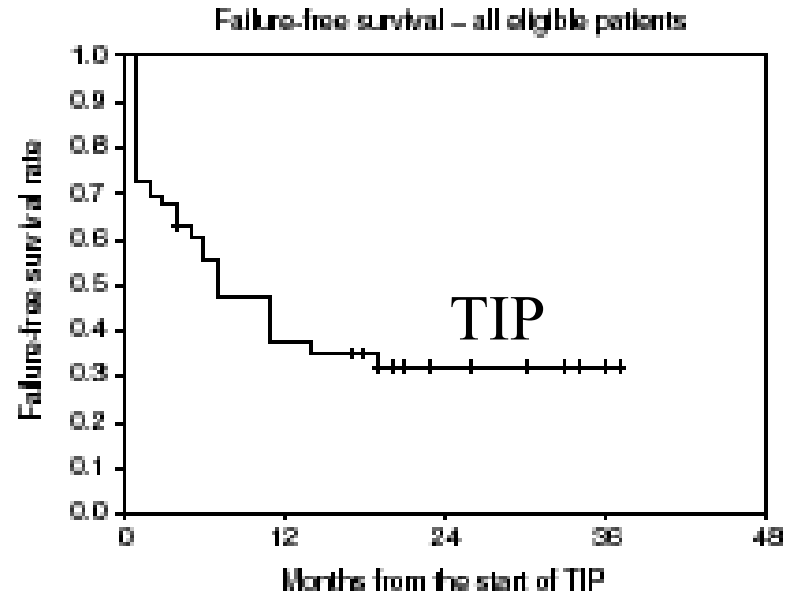


Fig 1. Progression-free survival for all eligible patients.



Mead et al 2005 for MRC

135 patients treated
1984-1989

43 patients FFS at 1 year 38%

About 30-40% salvage overall, higher in seminoma and if patients are selected

IT 94--Randomised trial of high dose vs conventional dose salvage

Overall response = 59% vs 61%(HD)

EFS at 3 years 35% vs 42% (HD) $p=0.16$

OS at 3 years 51% in both arms

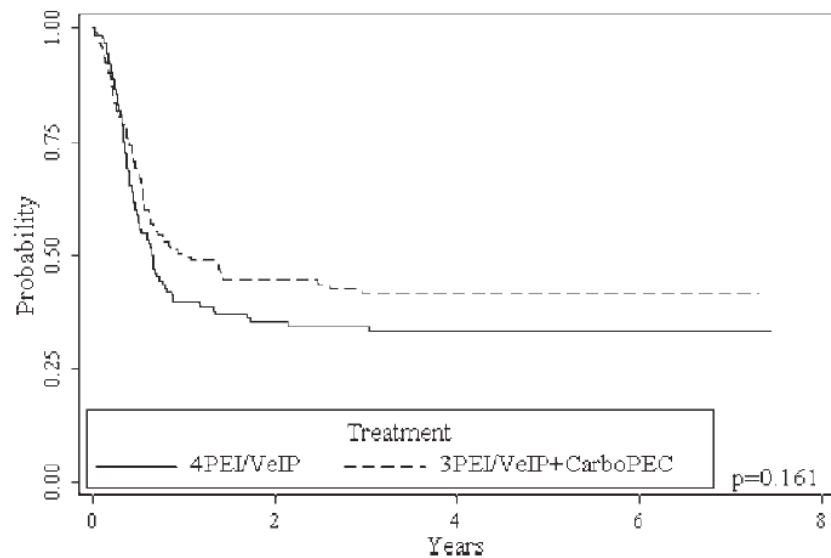


Figure 1. Event-free survival.

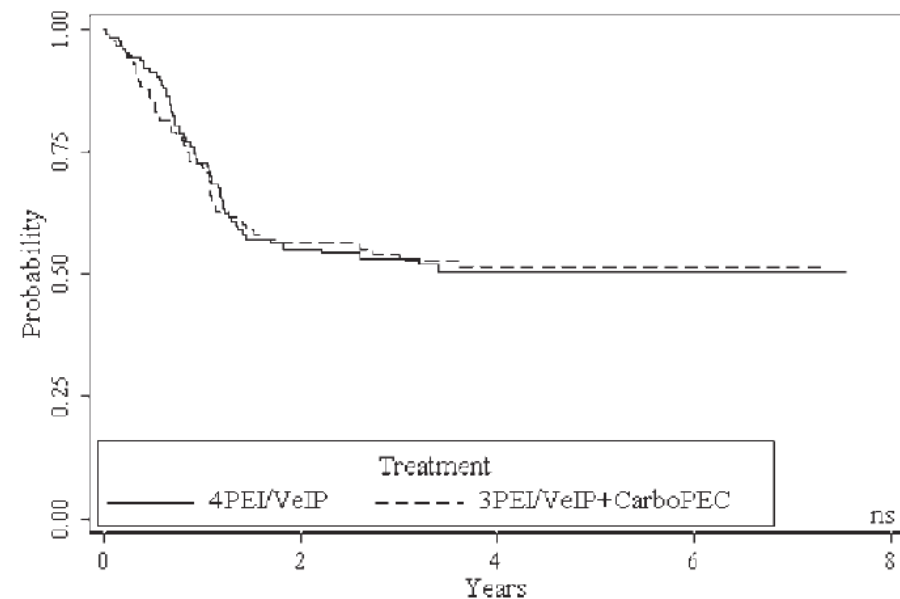


Figure 3. Overall survival.

Tandem HDC after salvage induction

Einhorn 07 NEJM

184 patients progressing after Cisplat cc

63% DFS

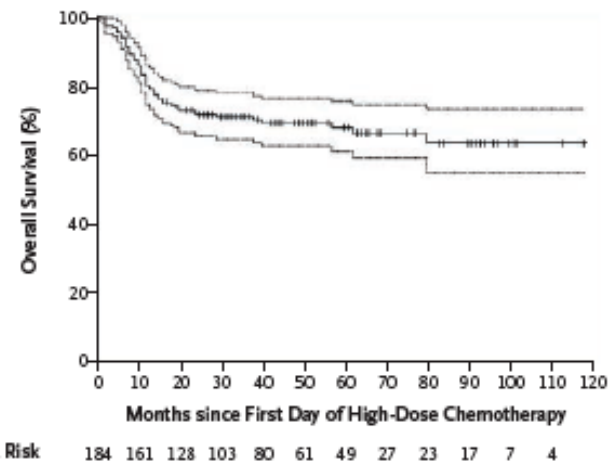


Figure 1. Kaplan-Meier Estimates of Overall Survival.

The top and bottom lines show the 95% confidence interval.

Similar results with
TI-CE from MSKCC
(3xHDC)

TT management summary 2014

Stage I seminoma -----Surveillance or Carboplatin x 1

Stage I nonseminoma-----Surveillance or BEP x 1

Good prognosis -----BEPx3

Intermediate-----BEPx4

Poor prognosis-----BEPx4 Salvage-----
-----TIPx4 (Tandem high dose)

PLUS

Surgery to residual masses after chemo for nonseminoma